

DVCS Oracle and the Task of Copyright Preservation in Blockchain-Based Technology

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Abstract— the authors address the problem of copyright registration. The authors show the shortcomings of existing copyright confirmation systems based on blockchain platforms, and provide solutions that eliminate these shortcomings. A distinctive feature of the proposed solution is using a bundle of a blockchain platform with an input DVCS Oracle based on the Litoria DVCS hardware-software complex used to perform the copyright registration function. The blockchain platform is entrusted with the classic task of preserving documents confirming copyright. Authors conduct a smart contract in the Solidity language, which performs the creation of a document and its issuance at the user's request.

Keywords— *DVCS Oracle, hardware-software complex Litoria DVCS, CPD request and receipt, smart contract, copyright*

I. INTRODUCTION

In the era of digitalization of the economy and everyday life, each of us somehow creates copyright objects: programs, articles, photos for Instagram, videos for youtube. The rights to such objects (copyright objects) belong to the authors by default.

Copyright objects include books, films, music, fine art, sculpture, computer programs, databases, drawings, maps, articles.

The authors have two types of rights — exclusive (property) and non-property. Exclusive rights allow authors to prohibit the use of works by other persons.

The Berne Convention for the Protection of Literary and Artistic Works, adopted in 1883, established an international standard for protecting authors' rights [1]. To date, the document has been ratified in 176 countries.

The Berne Convention states that there is no need to register copyright objects: the document automatically protects them immediately after creation — no additional formalities are required.

However, the absence of registration requirements can also play against the author — no one is immune from disputes about the original authorship. As a rule, authors and companies voluntarily take measures that will help confirm authorship in the event of a dispute.

What can be done:

- Send the manuscript (disk, flash drive with the work) to yourself by mail.
- To deposit a copy of the work with a notary. Such a service varies; the average price is \$ 60 and above.
- Register copyright with government depositors: in the USA through the Copyright Office or in Canada on a particular portal of public services. The prices and terms of registration vary, but the benefit will not cost less than \$ 50. The reader can be found additional information at the link [2].
- Register copyright with private depositors, for example, with a respectable large depository INTEROCO Copyright Office in Germany. The cost of the service is 200 euros.

In Russia, you can join private domestic depositories. The cost of services for depositing the work - from one hundred rubles. However, no one can guarantee that the organization will last for a long time and will be able to provide proof in 20 years.

An alternative way to confirm copyright is to save the work in a decentralized public registry - in the blockchain. This method:

- The cheapest.
- Has the same legal force as the above.
- Does not force you to rely on the viability of any organization.

Similar services already exist Proof of Existence [3], Emernotar.io [4], a joint solution of Bitfury and PatentBot [5], or the Depositor service [6]. They all work the same way: the uploaded file is hashed [7], the result (a unique fingerprint of the file - hash) is entered into the blockchain. If verification is necessary, a repeated hashing operation is performed, and the resulting hash is compared with the hash stored in the blockchain. A hash match ensures that a specific file was entered into the blockchain at a particular time.

If it is necessary to use proof, the author who made the entry in the public blockchain must prepare the wording for the claim or claim and repeat the hashing operation.

The result of repeated hashing must match the hash recorded in the blockchain. After the operation, it remains only to explain to the court what hashing is and how the blockchain works.

Unique tokens are used - a digital cryptographic certificate of the object to confirm ownership of digital assets and the right to use them on several blockchain platforms. Such an object can be works of art, electronic documents, digital photographs, and other things of copyright. It is worth noting that such tokenization does not exclude copying objects but only secures ownership rights to them.

The most popular platform for creating tokens has become a decentralized platform based on the Ethereum blockchain. The key feature of this platform is the support of smart contracts executed in a single decentralized virtual machine EVM [8]. Non-interchangeable ERC-721 [9] tokens in Ethereum smart contracts can be used to build a decentralized platform for confirming the authorship of documents.

However, most of the listed systems have inconveniences for their use. If you have used any of these services, you could face some disadvantages:

- Lack of opportunities to issue and preserve documents (certificates) confirming ownership of the copyright object and suitable for applying to the court.
- State accreditation of a hardware-software complex and a legal entity as a trusted third party (TTP).
- Use of certified means of electronic signature (ES), time stamps (TSP) [10], and online verification of the status of certificates (OCSF) [11].
- Correct legal consolidation of various facts during the life cycle of the copyright object, including its transfer to another owner.

These shortcomings lead to the fact that the results of authorship confirmation can be considered null and void for the court.

According to the authors, all these shortcomings result from improper use of the blockchain platform to create authorship confirmation documents. This function is not peculiar to the blockchain platform, designed mainly for long-term and reliable storage of various documents in a public distributed registry, which the blockchain perfectly copes. The process of creating such documents, according to the authors, should be isolated and organized on a separate device working in conjunction with the blockchain platform.

In this regard, the authors propose the idea of separating the functions of creating copyright documents and storing them.

II. FROM THE IDEA TO THE SOLUTION OF THE PROBLEM OF COPYRIGHT CONFIRMATION

We will begin the presentation of the idea with an example of copyright registration for a photograph of Michelangelo's fresco "Adam's Creation," shown in Figure 1.

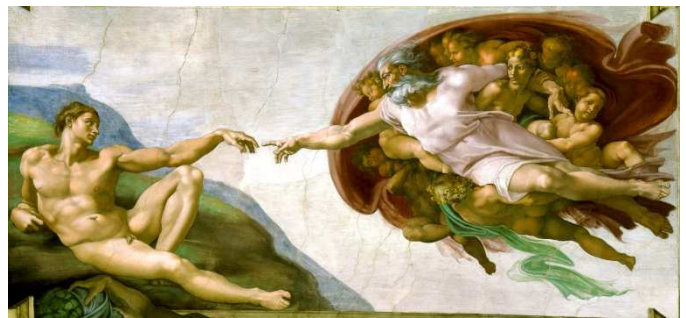


Fig. 1. A photograph of Michelangelo's fresco "Adam's Creation."

The structural diagram of the copyright registration system project is shown in Figure 2. The main components of the structural scheme of the copyright registration system project are:

1. The user's automated workplace (AWP) is based on the Litoria Desktop software package.
2. Litoria DVCS Oracle Server based on the hardware-software complex (HSC) Litoria DVCS Oracle.
3. Blockchain platform.

We will briefly describe the process of functioning of the system. The process begins with the preparation by the user of a CPD request [12] for registration using the Litoria DVCS software package of copyright for a photograph.

The input parameter is a photograph as an object of copyright. The output parameter is a CPD request, which is sent along with the photo to the input of the HSC Litoria DVCS Oracle server.

As a result of processing the CPD request, HSC Litoria DVCS Oracle generates a receipt (certificate of ownership of the copyright object) in the form of a so-called CPD receipt, shown in Figure 3.

This certificate displays the following information:

- The final status of CPD request processing;
- Parameters of the electronic signature (ES) of the Oracle DVCS server:
 - confirmation of data integrity,
 - validity of the ES verification key;

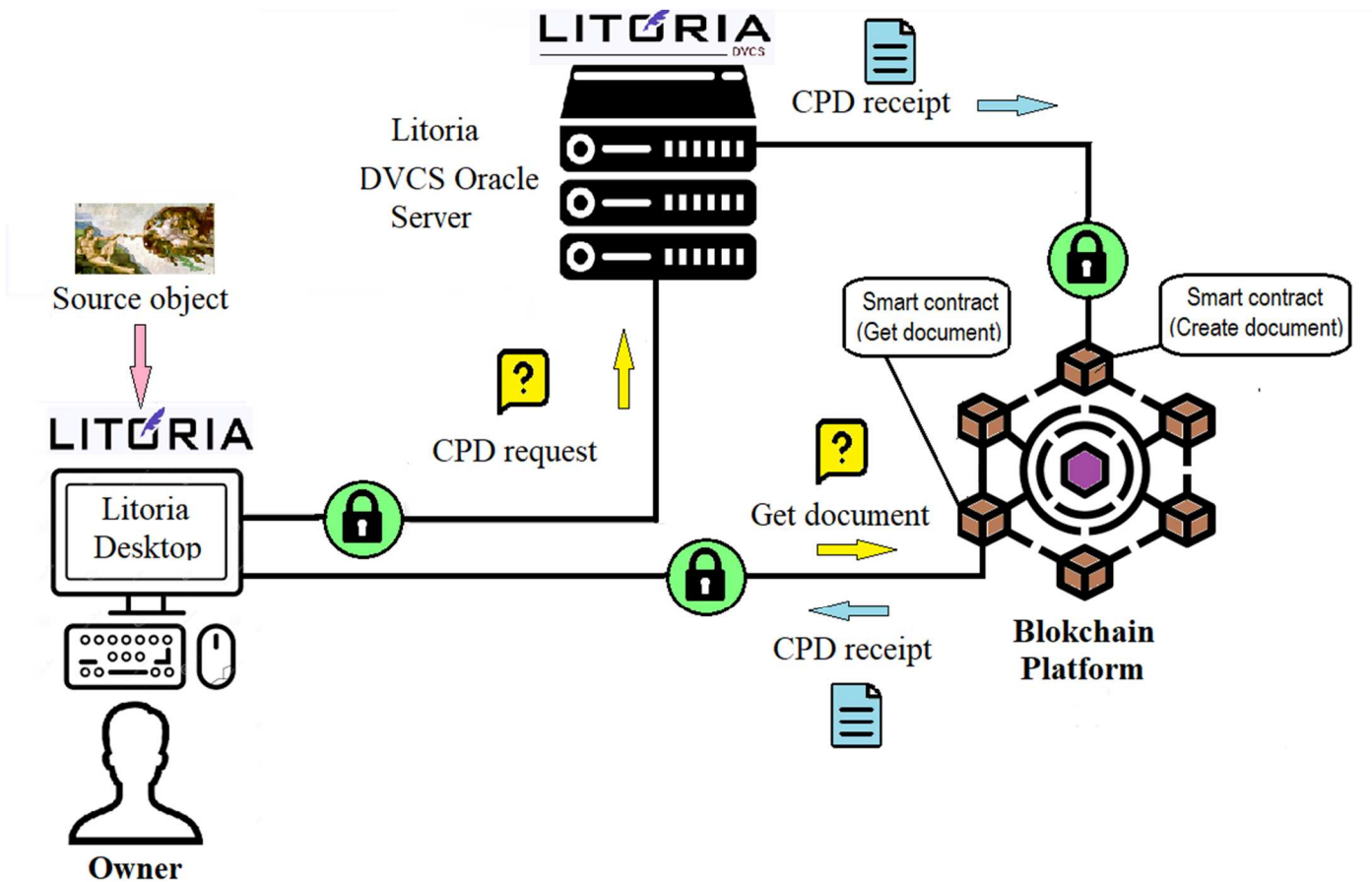


Fig. 2. Structural diagram of the project of the copyright registration system

- type ES;
- Date and time of the creation of the Oracle DVCS ES server.
- Parameters of the Oracle DVCS Server ES Verification Key Certificate:
 - name of the certificate owner;
 - name of the certificate publisher;
 - type of ES algorithm;
 - serial number of the certificate;
 - date and time of certificate creation.
- Receipt information:
 - Type of DVCS request;
 - date and time of request creation;
 - date and time of receipt creation.
- Hash stamp parameters:

- hash value;
- hashing algorithm.

The prepared CPD receipt enters the blockchain platform. It is processed using the "create document" function of the corresponding smart contract, described below, and stored in long-term public memory as a character string for further use.

Suppose it is necessary to extract the CPD receipt from the blockchain memory. In that case, the user can create a transaction with a call to the "get document" function at the smart contract's address and load the CPD receipt in the memory of the user's AWP.

Thus, the project proposed by the authors eliminates all the previously noted shortcomings:

- The project provides the issuance of a document confirming the ownership of the copyright object at the time of accessing the service;

- The functions of DVCS Oracle are performed by a particular state-accredited organization LLC UC GIS with high insurance coverage;
- Cryptographic means certified by the established procedure are used at all stages.



Fig. 3. Certificate of ownership of the copyright object

Next, we will look at the structure of HSC Litoria DVCS Oracle and a sample smart contract serving it.

III. MODULAR STRUCTURE OF HSC LITORIA DVCS ORACLE

The main task of HSC Litoria DVCS Oracle is to verify the authenticity of ES in electronic documents and the validity of ES key certificates during information exchange at a fixed time about the sender or recipient of an electronic document, as well as to confirm the ownership of information with or without providing it to the service (by hash).

The following main functions are implemented in HSC Litoria DVCS Oracle to perform this task:

1. Creation of the following types of DVCS requests (according to RFC3029):
- confirmation ES an electronic document (Validation of Digitally Signed Document – VSD);

- confirmation of the validity of the signature key certificate (Validation of Public Key Certificates – VPKC);
 - identification of the possession of information to the specified point-in-time delivery of its service (Certification of Possession of Data – CPD);
 - designation of information control without providing its services (via hash) (Certification of Claim of Possession of Data – CCPD).
2. Test data contained in the request.
 3. Formation of time stamps.
 4. Formation and analysis of DVC receipts by the recommendations of RFC3029.
 5. Archiving and storage of profile information in the database.
 6. Archiving and storage of DVC issued receipts and their serial numbers.
 7. Archiving and storing the event log files of the HSC Litoria DVCS Oracle authentication server containing the date, time, and type of the event being committed.

The modular structure of HSC Litoria DVCS Oracle is shown in Figure 4. Consider the composition and purpose of the components of the modular structure.

The DVCS Authentication service (the full description of the DVCS protocol, on which the service is based, is given in RFC3029 [13]) provides data verification services in the request received by the service, confirming the validity of the ES document, the validity period of the ES verification key certificates. The service also testifies to the provision of data and the fact that the user has this data at the specified time.

The DVCS Authentication Service (from now on referred to as the DVCS Service) includes the following components:

- DVCS web service;
- DVCS web portal;
- DVCS authentication service database.

The listed components of the service can function both on one server and in a distributed version on several servers.

The DVCS service allows you to:

1. Processing of all types of DVCS requests (according to RFC3029):
2. Formation of DVCS responses based on the data obtained during authentication.
3. Formation of DVC receipts by signing the DVCS response in the formats of the following standards:

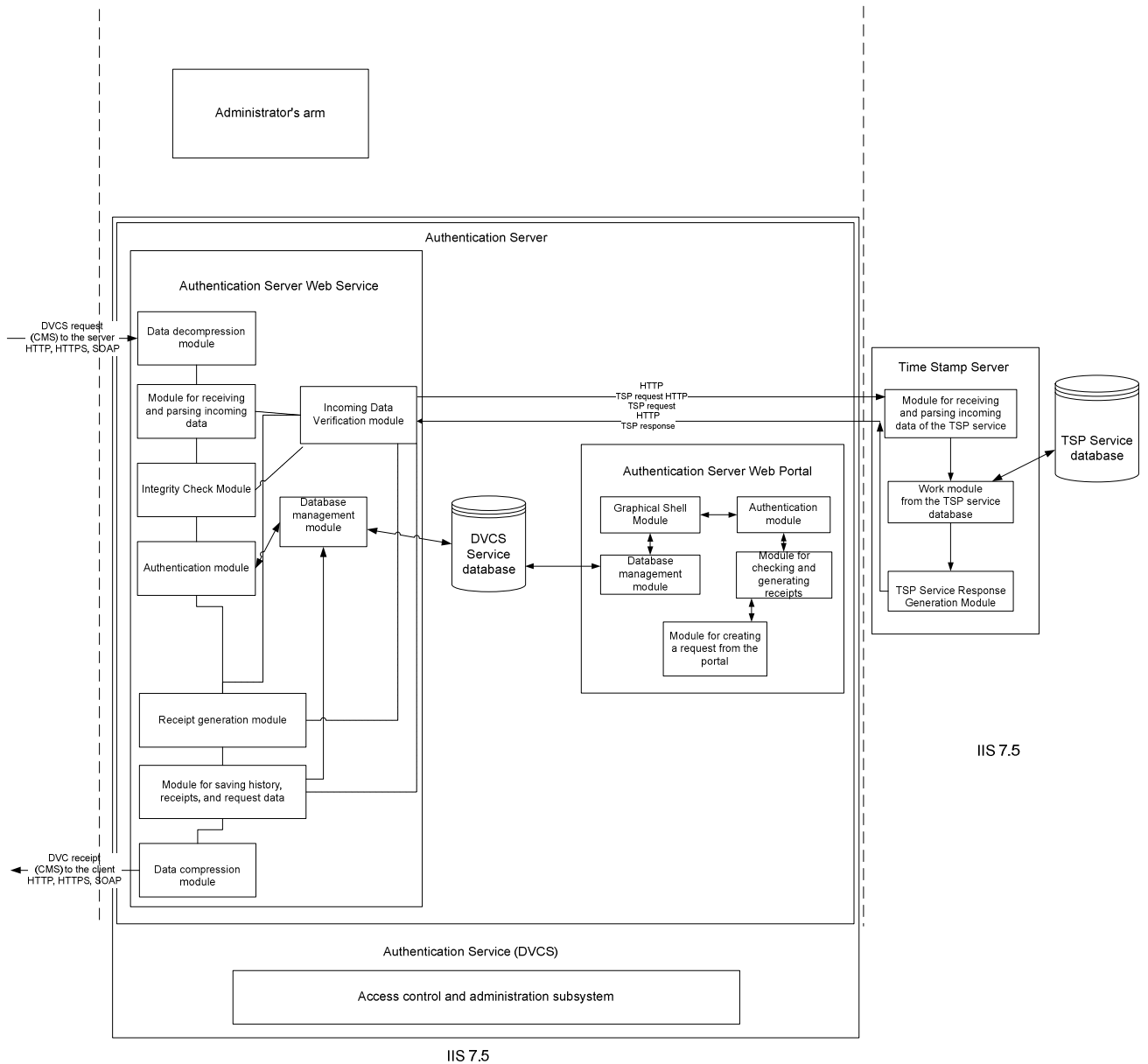


Fig. 4. Modular Structure of HSC DVCS Oracle

- a. CADES (RFC5126 «CMS Advanced Electronic Signatures (CADES)»);
- b. XAdES (ETSI TS 101 903 V1.4.1 «XML Advanced Electronic Signatures (XAdES)»).
4. Search and issue receipts for a previously made DVCS request.
5. Forming (sending) a request to receive a time stamp.

The DVCS web service runs under IIS 7.5/8.5. It uses the DVCS authentication service database running Microsoft

CSBD SQL Server to save the results of executed queries, authenticate users, and calculate statistical information.

The DVCS web service authenticates all requests to the service based on the certificate of the ES verification key with which the request is signed.

The interaction of users with HSC is carried out using the AWP of the Litoria Desktop user (see Figure 1) by sending signed DVCS requests to the DVCS web service. In response to the DVCS request, the service generates and sends a receipt with the results of the checks performed to the blockchain.

The database of the DVCS service running Microsoft SQL Server SCBD provides the ability to maintain/store the following data:

- profile information;
- DVC issued receipts and their serial numbers;
- HSC Litoria DVCS Oracle event log files containing date, time, and type of action;
- service information.

DVC issued receipts are stored in the DVCS service database indefinitely. The security of DVC receipts is guaranteed by secure authentication and SCBD mechanisms. It is not possible to delete receipts from the administration cabinet of the DVCS web portal.

Upon expiration of the validity period of the receipt verification, it is necessary to use the archive storage module "Litoria Arch Update" to extend the term of the trusted archive storage of signed receipts.

The DVCS web portal runs under IIS 7.5/8.5 and includes an administrator's office for access control and administration of the DVCS service.

Work with the portal is carried out using a web browser (with support for cryptography certified in Russia) using a secure HTTPS connection (with support for the TLS protocol) in a controlled zone.

The DVCS web portal authorizes the connection of administrators (the system administrator and the audit administrator have access to the web portal) using a TLS connection. Then, within the established relationship, it requests a login and password. If both types of authentication are successful, it allows the specified administrators to administer the HSC components through the administrator's cabinet.

The administrator's office for access control and administration of the DVCS service provides the ability to perform the following functions:

- configuring HSC Litoria DVCS Oracle;
- management of profiles and accounts accessing and having access to HSC Litoria DVCS Oracle (including separation of powers, control of authentication settings, and access rights);
- setting up mutual authentication of HSC Litoria DVCS Oracle servers among themselves;
- View audit log and system messages;
- search and receipt of receipts for previously made DVCS requests in CAdES/XAdES formats;
- create and send VSD, VPKC, CPD and CCPD requests to the Authentication Service (DVCS);
- receiving and verifying response receipts from the Authentication Service (DVCS).

Access to the administrator's office for access control and administration of the DVCS service is carried out remotely using an AWP administrator operating within the controlled zone through access through a web portal.

The TSP Time Stamp Service (the full description of the TSP protocol, based on which the work of the service is based), is given in RFC3161 [10]) certifies the existence of an electronic document at a certain point in time using timestamps. The TSP service allows the formation of timestamps when fixing the time of the creation of the DVCS response and when fixing the time of construction of the DVC receipt (signature of the DVCS response).

The TSP Time Stamp Service (from now on referred to as the TSP Service) consists of the following components:

- TSP web service;
- TSP web portal;
- TSP time Stamp service database.

The listed components of the service can function both on one server and in a distributed version on several servers (while the TSP web service and the TSP web portal always work on the same server).

With the help of a timestamp, the TSP service certifies that it has been provided with the value of the hash function of the document, the existence of which must be certified no later than the time specified in the time stamp. The hash function value itself is also indicated in the time stamp.

In any valid HSC Litoria DVCS Oracle configuration, there must be an administrator AWP with which the administrator will perform access to the HSC Litoria DVCS Oracle web portal. The administrator's AWP functions under Windows OS. Also, a web browser must be installed on this AWP that uses a certified means of cryptographic protection of information that provides TLS.

In the next section, the authors consider the program code of a smart contract that ensures the relationship of the user and HSC Litoria DVCS Oracle with the blockchain platform.

IV. SMART CONTRACT DEVELOPMENT

As already mentioned by the authors in section I, the ERC-721 standard is an open standard describing how to create non-interchangeable tokens on blockchains compatible with EVM. Using this standard and the Solidity smart contract description language, programmers can create new contracts compatible with ERC-721 by importing the implementation of the middle from the OpenZeppelin library, which the authors did.

For example, the authors have developed a smart contract to confirm the authorship of electronic documents, which uses HSC Litoria DVCS Oracle as a source of document verification results presented as a character string about 900 bytes long. The smart contract provides a function for writing a string to memory (createDocument) and a procedure for extracting it from memory (getDocument) at the request of the corresponding transactions.

The record function takes two arguments: the address of the document owner in the blockchain and a character string of the document verification results and returns the token ID in the blockchain. The reader function takes the token ID as an argument and returns a character string.

The listing of the smart contract program in the Solidity language is given below:

```
// SPDX-License-Identifier: UNLICENSED

pragma solidity >=0.7.0 <0.9.0;

import
'@openzeppelin/contracts/token/ERC721/ERC721.sol';
import '@openzeppelin/contracts/utils/Counters.sol';

contract ERC721DocumentStorage is ERC721 {
    using Counters for Counters.Counter;
    Counters.Counter private _tokenIds;
    mapping(uint256 => string) private
_tokenDocumentMapping;

    constructor () ERC721 ("DocumentStorage token",
"DST") {}

    function createDocument(address owner, string
memory tokenURI) public returns (uint256) {
        require(bytes(tokenURI).length == 900,
"Symbol string must have length 900 symbols");

        _tokenIds.increment();

        uint256 newDocumentToken =
_tokenIds.current();
        _safeMint(player, newDocumentToken);
        _setTokenDocument(newDocumentToken,
tokenURI);

        return newDocumentToken;
    }

    function getDocument(uint256 tokenId) public
view returns (string memory){
        require(!_exists(tokenId),
"ERC721DocumentStorage: nonexistent token");
        return _tokenDocumentMapping[tokenId];
    }

    function _setTokenDocument(uint256 tokenId, string
memory _documentSign) internal virtual {
        require(!_exists(tokenId),
"ERC721DocumentStorage: Document set of nonexistent
token");
        _tokenDocumentMapping[tokenId] =
_documentSign;
    }

    function _burn(uint256 tokenId) internal virtual
override {
        super._burn(tokenId);

        if
(bytes(_tokenDocumentMapping[tokenId]).length != 0)
{
            delete _tokenDocumentMapping[tokenId];
        }
    }
}
```

V. CONCLUSION

Thus, in this article, the authors solved the problem of confirming authorship of the copyright object based on the use of the blockchain platform in conjunction with HSC Litoria DVCS Oracle. This solution is new because it eliminates the significant shortcomings of existing copyright registration platforms using the blockchain platform. The blockchain bundle with HSC Litoria DVCS Oracle has broader capabilities than presented in this article. The authors show an example of using only one type of DVCS request, namely CPD. But the solution developed by the authors is universal and can be further used in all areas of electronic signature application for its correct verification. For example, the authors have successfully applied this solution to electronic document management systems in global supply chains [14].

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